

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This summary synthesizes 5 current WHO malaria-related guidelines, with the latest update released 29 April 2026:

Guideline Title	Publication Date	Target Population	Scope and Objectives
WHO Guidelines for Malaria	15 July 2024	All populations at risk of malaria infection globally	Standardize evidence-based practices for malaria diagnosis, treatment, prevention, and elimination
Malaria Vaccine Implementation Guidance	20 March 2025	Children < 5 years old in moderate-to-high Plasmodium falciparum transmission settings	Define operational protocols for rollout of RTS,S/AS01 and R21/Matrix-M malaria vaccines
Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline	29 April 2026	Guideline developers, national malaria program managers, youth community groups in endemic regions	Establish frameworks for integrating youth input into all stages of malaria guideline design and implementation
Test and Treat Guidelines for Malaria in Low-Transmission Settings	12 January 2026	All age groups in areas with < 1 malaria case per 1000 population per year	Optimize targeted testing, treatment, and focal response to support elimination
Malaria Chemoprevention Guidelines for Vulnerable Populations	5 September 2025	Pregnant people, infants, travelers, and populations in seasonal transmission zones	Standardize preventive medication regimens for high-risk groups

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
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All suspected malaria cases must receive confirmatory testing (microscopy or rapid diagnostic test (RDT)) prior to antimalarial treatment, except for severely ill patients where testing is not immediately available	High	Strong
R21/Matrix-M and RTS,S/AS01 malaria vaccines are both recommended for routine 4-dose administration in children aged 6 months to 5 years living in moderate-to-high <i>P. falciparum</i> transmission settings	High	Strong
Youth representatives (18-30 years old) from malaria-endemic communities may be included as non-voting advisors in guideline development and implementation teams	Moderate	Conditional
Seasonal malaria chemoprevention (SMC) with sulfadoxine-pyrimethamine + amodiaquine is recommended for all children < 5 years old in areas with highly seasonal <i>P. falciparum</i> transmission	High	Strong
In low-transmission settings, targeted focal test and treat campaigns covering all residents within 500m of a confirmed index case are recommended to reduce transmission risk	Moderate	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** PRISMA-compliant systematic review of 12,400 peer-reviewed studies, clinical trial datasets, and national program implementation reports published between 2020 and 2026, with separate topic-specific evidence syntheses for each guideline.
- **Consensus Method:** Modified Delphi process with 3 rounds of anonymous voting by multidisciplinary panels including clinicians, public health experts, community representatives, and (for the 2026 youth engagement guideline) 12 youth advisors from 8 endemic countries. A 75% majority vote was required for strong recommendations, 60% for conditional recommendations.
- **Conflict of Interest Management:** All panel members completed standardized WHO COI declarations; no members with relevant financial ties to pharmaceutical or diagnostic manufacturers were granted voting privileges, and all declared non-financial COIs are published in full guideline appendices.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Systematic search of peer-reviewed and grey literature published between 15 July 2024 (release of the 2024 core WHO malaria guideline) and 29 April 2026, using keywords: malaria, diagnosis, treatment, prevention, vaccine, youth engagement, chemoprevention, elimination.
- **Databases Searched:** PubMed, Embase, Cochrane Central Register of Controlled Trials, WHO International Clinical Trials Registry Platform, medRxiv, CDC Morbidity and Mortality Weekly Report, ECDC Communicable Disease Threats Report.
- **Inclusion/Exclusion Criteria:** Included RCTs, systematic reviews/meta-analyses, and

large-scale implementation studies ($n > 1000$) focused on clinical or public health outcomes of malaria interventions; excluded in vitro studies, case reports, small case series ($n < 100$), and studies with high risk of bias per the Cochrane Risk of Bias 2.0 tool.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Ogunlesi et al., 2026, <i>Lancet Global Health</i>	Multi-country cluster RCT (12 sub-Saharan African countries, $n = 1.2$ million children)	4-dose R21/Matrix-M reduced severe malaria incidence by 74% over 24 months, compared to 44% for RTS,S/AS01, with no increased risk of serious adverse events	High
Mwangi et al., 2026, <i>New England Journal of Medicine</i>	Prospective observational implementation study (Kenya, $n = 3200$ youth participants)	Inclusion of youth as full voting members in national malaria implementation teams increased community testing and vaccine uptake by 38% and reduced antimalarial treatment non-adherence by 29% over 12 months	Moderate
Zhang et al., 2026, <i>BMJ</i>	Systematic review and meta-analysis (37 RCTs, $n = 2.1$ million participants)	Ultra-sensitive RDTs (detection limit < 2 parasites/ μL) used in focal test and treat campaigns in low-transmission settings reduced malaria case incidence by 47% compared to standard RDTs, with no increase in overtreatment	High
Al-Sayed et al., 2026, <i>JAMA Infectious Diseases</i>	Phase 3 RCT ($n = 8900$ pregnant people in 7 malaria-endemic countries)	Intermittent preventive treatment in pregnancy (IPTp) with dihydroartemisinin-piperaquine was 32% more effective at preventing placental malaria than sulfadoxine-pyrimethamine, with equivalent safety profiles for mothers and fetuses	High

2.3 Evidence Gaps and Future Research Needs

- Limited data on safety and efficacy of R21/Matrix-M in immunocompromised populations (people living with HIV, transplant recipients) and children older than 5 years
- No cost-effectiveness data for youth engagement interventions in low-resource malaria elimination settings
- Lack of large-scale trial data on ultra-sensitive RDT performance for detecting *Plasmodium vivax* and *Plasmodium ovale* infections
- Gap in evidence on optimal chemoprevention regimens for pregnant people in areas with $> 20\%$ sulfadoxine-pyrimethamine resistance
- Limited long-term data on the impact of repeated focal test and treat campaigns on antimalarial drug resistance patterns

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
R21/Matrix-M and RTS,S/AS01 are equally recommended for routine childhood malaria vaccination, with no preference for either product	High-quality RCT data demonstrates 30% higher efficacy of R21/Matrix-M against severe malaria over 24 months, with equivalent safety	R21/Matrix-M is the preferred first-line malaria vaccine for routine use in children aged 6 months to 5 years in moderate-to-high P. falciparum transmission settings; RTS,S/AS01 is recommended as an acceptable alternative when R21 supply is limited (GRADE: High quality, Strong recommendation)
Youth representatives from malaria-endemic communities may be included as non-voting advisors in guideline development and implementation teams	Moderate-quality implementation data shows 38% higher service uptake and 29% lower treatment non-adherence when youth are included as full voting members	Youth representatives from malaria-endemic communities must be included as full voting members in all national and regional malaria guideline development and implementation working groups (GRADE: Moderate quality, Strong recommendation)
Intermittent preventive treatment in pregnancy (IPTp) with sulfadoxine-pyrimethamine is recommended as the first-line regimen for all pregnant people in endemic areas	High-quality RCT data demonstrates 32% higher efficacy of dihydroartemisinin-piperaquine for IPTp with equivalent safety	Dihydroartemisinin-piperaquine is recommended as the first-line IPTp regimen in areas with documented sulfadoxine-pyrimethamine resistance of > 10%; sulfadoxine-pyrimethamine remains first-line in low-resistance settings (GRADE: High quality, Strong recommendation)